

National University of Computer and Emerging Sciences



Lab Manual 10 **Computer Organization and Assembly Language Lab**

Department of Computer Science
FAST-NU, Lahore, Pakistan

Lab Manual – TSRs and Interrupts

Exercise 1: Your task is to get two keystrokes using keyboard interrupt number 0x16 one after the other. On the first keystroke, regardless of which key is pressed, you should clear the screen.

The following should be done on second keystroke:

- If you press 'a', your program should display "Hi, You pressed a".
- If you press 'b', your program should display "Hi, You pressed b".
- For any other case your program should display "Hi, You entered wrong credentials".

After two keystrokes, your program should terminate.

Exercise 2: Write a TSR and which manipulates the given string and then shows it onto the screen according to given rules:

- While left shift is pressed, the word "and" present in string must be replaced by "dna" and whole string is to be displayed onto screen.
- If left shift is released, your screen should be cleared
- While right shift is pressed, the word "has" present in string must be replaced by "sah" and whole string is to be displayed onto screen.
- If right shift is released, your screen should be cleared
- For any other buttons, your code should not do anything and pass the control to Original ISR designed for keyboard interrupts

Scan Code (Left Shift Press): 0x2a

Scan Code (Right Shift Press): 0x36

Example:

String: db 'He has food and drinks',0

On Left Shift Press: He has food dna drinks

On Right Shift Press: He sah food and drinks